Execution graph	**Stage 2:** program	**Stage 3:** gem5 harness
Store bypass i0 `Store` ┐*rf* i1 `Load` ◄┘ *unresolved*	i0 and i1 share an address; inline asm, so no compiler forwarding.	i0's address held 5000 cycles; i1 forced to predict no alias.
Unresolved branch, mispredicted either way i0 `Branch_n` *unresolved*	Constant condition: exits architecturally, so the rest runs only speculatively.	Forced to mispredict; resolution held 5000 cycles; still unresolved at the xmit.
Resolved branch, not taken i0 `Branch_n` *resolved*	Zeroed condition: falls through, never taken.	Forced correct; resolution delayed 0, 500 or 2500 cycles (all combinations).
Commit boundary i0 lc *committed* i1 fnc *uncommitted*	Global labels on lc and fnc (Alloy instructions only).	Window: lc retired < xmit < fnc retires (measured); no commit stall.
Target transmitter i0 `Branch_x` *xmit* *or* `Other_x` *xmit*	`Branch_x`: `testq+jz` on its operand, architecturally taken. Divide: unguarded, so a zero divisor faults.	`Branch_x`: forced to mispredict, no hold. Divide: a fault after a mispredict is a transmission.